Research Article Vol. 34, No. 8 / 20 Apr 2026 / Optics Express 14943
Optics EXPRESS

Doubled Z-axis scanning range and spot-quality enhancement in system-level integrated co-optimized transmissive 3D laser scanning systems

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Abstract: Transmissive three-dimensional (3D) laser scanning systems are promising for precision micro- and nanofabrication, but their scanning range is limited by cumulative optical path difference (OPD) induced by large-field angular scanning and dynamic axial focusing. We propose a system-level integrated co-optimization (SLICO) strategy in which the dynamic focusing module, XY galvanometric scanner, and F-theta telecentric lens are jointly modeled and optimized for collaborative aberration compensation as a unified system. Simulations demonstrate that, over an 80 mm diameter XY scanning field and with the Z-axis scanning range extended from ±10 mm to ±20 mm, the OPD is reduced from 1.0 to 0.25 waves and the Strehl ratio (STRH) rises from 0.3 to above 0.96. Experimental validation confirms the strategy's feasibility and performance. This co-optimization strategy significantly enlarges the scanning field for a high-quality focal spot and provides a practical solution for high-precision, large-range laser fabrication.

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1. Introduction:

Laser 3D scanning technology, which relies on the rapid movement of the laser focus in the 3D space, has found widespread applications in laser material processing [1–4], laser cleaning [5–8], and 3D printing [9–12]. In such systems, the coupled operation of the dynamic focusing module, XY galvanometer scanner, and F-theta telecentric lens plays a decisive role in determining the focal spot quality and the effective scanning range, thereby directly influencing the overall processing efficiency and precision. However, in transmissive 3D laser scanning systems, these optical components are generally designed and optimized independently, with each module minimizing its own optical path difference (OPD). When the modules are integrated into a complete system, optical coupling among the subsystems introduces additional system-level aberrations and OPD that are not accounted for in the separate designs. Moreover, the lateral beam deflection induced by the XY galvanometer scanner further amplifies the OPD across the scanning field. As a result, the focal spot quality progressively degrades along both the axial and lateral directions. This degradation becomes particularly pronounced over large scanning volumes, where the STRH, focal spot circularity and diameter deteriorate significantly, leading to reduced processing uniformity and a restricted scanning range.

In response to the growing demand for high-speed and high-precision 3D laser processing in both industrial and scientific applications, various advanced Z-axis dynamic focusing elements have been developed, including variable-focus liquid lenses [13–18], tunable acoustic gradient-index (TAG) lenses [19–22], and deformable mirrors [23–26]. These devices offer advantages

#588979 https://doi.org/10.1364/OE.588979
Journal © 2026 Received 7 Jan 2026; revised 21 Feb 2026; accepted 1 Apr 2026; published 13 Apr 2026

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in terms of response speed, focal tunability, and system compactness. Despite these advances, under large scanning ranges and high-speed dynamic focusing conditions, residual aberrations and focal spot distortions [27,28] remain significant challenges. In addition, reflective dynamic focusing devices achieve 3D laser processing by converting galvanometer angular motion into axial displacement [29–31]. However, such systems typically require four galvanometer scanners, which increases system cost and complexity while restricting the achievable 3D scanning range. As a result, achieving uniform and high-quality focal spots over an extended 3D scanning range remains challenging for existing dynamic focusing schemes, thereby limiting the feasibility of large-area, high-precision 3D laser processing and motivating further investigation into transmissive 3D laser scanning systems. In contrast, studies on transmissive 3D laser scanning systems remain relatively limited, and the interplay between system configuration and aberration remains insufficiently understood with a lack of systematic design methodologies.

In this paper, the mechanisms that limit the scanning range in transmissive 3D laser scanning systems are systematically investigated. A system-level integrated co-optimization (SLICO) strategy is proposed, in which the dynamic focusing module, XY galvanometer scanner, and F-theta telecentric lens are jointly optimized to suppress system-level OPD and expand the usable 3D scanning range. A comprehensive simulation model is developed to predict the evolution of the STRH, focal spot circularity and diameter throughout the entire scanning space. Experimental validation is subsequently performed. The results demonstrate that the proposed strategy maintains uniform, high-quality focal spots across the full 3D scanning range, thereby significantly improving processing consistency and reliability.

2. Simulation comparison of the conventional transmissive 3D laser scanning system and the optimized design

2.1. Description of transmissive 3D laser scanning system

In the proposed transmissive 3D laser scanning system, the dynamic focusing module functions as the Z-axis dynamic focusing module, and its optical configuration is illustrated in Fig. 1(a). The theoretical model comprises a movable negative lens P_1 and a fixed positive lens P_2. Specifically, axial translation of P_1 modifies in the effective focal length of the laser system, thereby enabling continuous adjustment of the focal position along the Z axial direction. In the optical layout, d_1, d_2 and d_3 represent the distances between P_1 and P_2, between P_2 and the ideal F-theta telecentric lens P_3, and between P_3 and the image plane (i.e., the working distance), respectively, while D represents the beam aperture of the system. Based on imaging theory, the functional relationship among the working distance d_3, the effective focal length f of the transmissive 3D laser scanning system, and the output beam divergence angle α of the dynamic focusing module can be derived. The corresponding mathematical expression is provided in Eq. (1). This model provides a theoretical basis for predicting the Z-axis focal shift and for optimizing the dynamic focusing performance of the scanning system.

$$\begin{aligned} d_3 &= f_3 \frac{f_1f_2 - d_1f_2 - d_2f_1 + d_1d_2 - d_2f_2}{f_1f_2 - d_1f_2 - d_2f_1 + d_1d_2 - d_2f_2 + f_1f_3 - d_1f_3 + f_2f_3} \\ f &= \frac{f_1f_2f_3}{f_1f_2 - d_1f_2 - d_2f_1 + d_1d_2 - d_2f_2 + f_1f_3 - d_1f_3 + f_2f_3} \\ \alpha &= \arctan\left\{\frac{D}{2}\left[\frac{1}{f_1} + \frac{1}{f_2}\left(1 - \frac{d_1}{f_1}\right)\right]\right\} \end{aligned} \tag{1}$$

To validate the accuracy of Eq. (1) in predicting the dynamic behavior of the system and to evaluate its dynamic focusing performance, numerical simulations were conducted using a custom-developed program. The theoretical values for the dynamic focusing performance were obtained from Eq. (1) and Eq. (2). For comparison, Zemax simulations were also performed to

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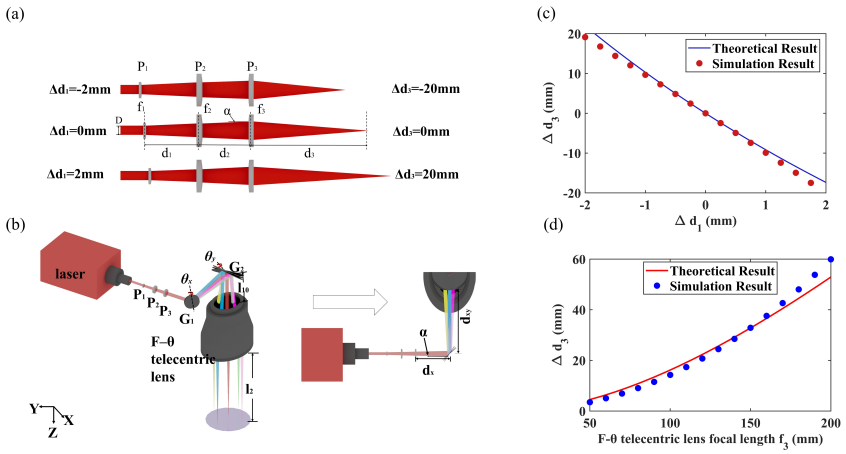

Fig. 1. (a) Schematic principle of the dynamic focusing module. (b) Optical layout of the transmissive 3D laser scanning system. (c) Analytical vs. simulated dependence of the focal shift Δd_3 on the lens displacement Δd_1. (d) Comparison between theoretical and simulation results for the maximum working distance shift under a ± 2 mm variation in d_1 across different F-theta lens focal lengths. Solid lines and scatter points denote the theoretical derivations and Zemax simulations, respectively.

generate the corresponding simulation results. The simulations employed lens P_1 with a focal length of f_1 = -26.6 mm, a positive lens P_2 with f_2 = 45.5 mm, and an ideal F-theta telecentric lens P_3 with f_3 = 167 mm. To compare the theoretical results from Eq. (1) with the simulation results, we conducted a side-by-side comparison, as shown in Fig. 1(c). While the theoretical predictions (based on Eq. (1)) and the simulation results exhibit a similar linear dependence of the working distance d_3 on the axial displacement of the negative lens P_1, slight differences can be observed. Specifically, when P_1 was translated by ±2 mm along the optical axis, the working distance d_3 varied by ±20 mm in theory, while the simulation results indicated a small deviation from this predicted value. This demonstrates that the system achieves a tenfold Z-axis dynamic focusing via an optical lever amplification mechanism, with both the theory and simulations showing consistent but slightly differing results. To validate Eq. (2) and compare it with simulation results, we include both theoretical predictions and simulations in Fig. 1(d). As shown in the figure, both the theoretical model and the simulation results reveal that the Z-axis dynamic focusing capability of the system increases with the focal length of the F-theta telecentric lens within a certain range. Although there are minor discrepancies between the theory and simulations, the trends in the data are in good agreement. This suggests that properly selecting the F-theta telecentric lens parameters can further enhance the achievable dynamic focusing range, with the theory and simulations showing good overall consistency.

The geometric optical model of the transmissive 3D laser scanning system is illustrated in Fig. 1(b). The system primarily consists of a dynamic focusing module, an XY galvanometer scanner, and an F-theta telecentric lens. The X and Y galvanometer mirrors, denoted as G_1 and G_2, rotate about their respective axes, enabling high speed beam scanning within the XY plane. The dynamic focusing module consists of three optical lenses, labeled P_1, P_2, and P_3. Collimated light first propagates through this module and is subsequently deflected by the galvanometer scanner, allowing precise positioning of the focal spot at any point P(x,y,z) in 3D space. Here, θ_x and θ_y denote the deflection angles of the X and Y galvanometers, respectively; h represents the output beam radius of the dynamic focusing module; α is the beam divergence angle. The distances d_x, d_{xy}, and l_{10} correspond to the spacing between the dynamic focusing module and the

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X galvanometer axis, between the X and Y galvanometer axes, and between the Y galvanometer axis and the F-theta telecentric lens, respectively, while f_1 is the focal length of the F-theta telecentric lens. After passing through the transmissive F-theta telecentric lens, the imaging distance is defined as l_2. Based on paraxial imaging theory [32], the positional coordinates of the system focal point P(x,y,z) can be calculated using the following relationship:

$$\begin{aligned} x &= (\sqrt{z^2 + y^2} + d_1)\tan 2\theta_x \\ y &= z\tan 2\theta_y \\ z &= l_{10} + l_{20} \end{aligned} \tag{2}$$

where,

$$\begin{aligned} l_1 &= \frac{l_{10}}{\cos 2\theta_y} \\ l_{20} &= l_2 \cos 2\theta_y \\ l_2 &= \frac{\frac{BG\cos\alpha}{\cos(2\theta_y-\alpha)} + l_1\tan(2\theta_y-\alpha) + \frac{BC\cos\alpha}{\cos(2\theta_y+\alpha)} + l_1\tan(2\theta_y+\alpha)}{\tan(2\theta_y+\alpha) - \tan(2\theta_y-\alpha) + \frac{1}{f_1}\left(\frac{BG\cos\alpha}{\cos(2\theta_y-\alpha)} + l_1\tan(2\theta_y-\alpha) + \frac{BC\cos\alpha}{\cos(2\theta_y+\alpha)} + l_1\tan(2\theta_y+\alpha)\right)} \\ BC &= BG = h - d_x\tan\alpha - \frac{d_{xy}}{\cos 2\theta_x}\tan\alpha \end{aligned} \tag{3}$$

As indicated in Eq. (2), the system focal position P(x,y,z) is determined jointly by the deflection angles of the XY galvanometer scanner and the divergence angle of the output beam emitted from the dynamic focusing module. The dynamic focusing module is actuated by a voice coil motor and, in combination with an optical lever amplification mechanism, exhibits dynamic response characteristics and Z-axis scanning speeds that are compatible with those of the XY galvanometer scanner. This configuration ensures high speed, synchronized 3D focusing. Because both the galvanometer deflection angles and the output beam divergence can be precisely controlled via input voltages, the laser focal spot can be scanned along a predetermined trajectory in 3D space by appropriately designing the voltage drive functions, while maintaining the system's dynamic performance under high-speed operation.

In the proposed 3D scanning system, the incident laser beam can be modeled as a monochromatic scalar wave. The electric field distribution E(r, t) at the entrance pupil is governed by the scalar wave equation:

$$E(r,t) = A(r)\exp[i(k\cdot r - \omega t + \phi)] \tag{4}$$

where $A(r)$ is the amplitude, k is the wave vector, w is the angular frequency, and φ represents the phase component. The phase component is intrinsically related to the wavefront function $W(\rho,\theta)$ as follows:

$$\phi = \frac{2\pi}{\lambda}W(\rho,\theta) \tag{5}$$

where ρ denotes the normalized pupil coordinate and θ is the azimuthal angle. In conventional transmissive 3D laser scanning systems, the aberration correction and the design of focal-plane flatness for the F-theta telecentric lens are typically performed under the assumption that the beam incident on the entrance pupil constitutes an ideal plane wave, where the wavefront function is constant:

$$W(\rho,\theta) = \text{const} \tag{6}$$

Under this condition, the phase distribution across the aperture is uniform, allowing system aberrations, including spherical aberration and coma, to be effectively compensated.

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However, the introduction of the dynamic focusing module induces a dynamic divergence angle α into the incident beam, transforming the wavefront from a plane wave into a spherical wave. This modifies the entrance-pupil wavefront function as follows:

$$W'(\rho, \theta) = W(\rho, \theta) + \Delta\phi(\alpha, \rho) \tag{7}$$

where $\Delta\phi(\alpha, \rho)$ represents the phase shift introduced by the beam divergence. Consequently, the phase distribution deviates from the ideal planar state, disrupting the original aberration balance. As a result, the pre-designed correction of the F-theta telecentric lens becomes ineffective, leading to focal spot distortion, increased spot diameter, and degraded 3D scanning performance.

2.2. Simulation and analysis of the conventional transmissive 3D laser scanning system

Based on the theoretical analysis, a conventional transmissive 3D laser scanning model with direct integration was constructed to further investigate the underlying mechanisms responsible for the observed performance degradation. To establish a rigorous comparative baseline in the optical simulations, the conventional 3D scanning system was modeled using an equivalent optical approach. Because the exact internal prescriptions of the commercial LINOS F-theta lens and the Scanlab dynamic focusing unit are proprietary, an equivalent F-theta lens model matching the macroscopic specifications of the physical setup (focal length $f = 167$ mm, operating wavelength $\lambda = 1064$ nm) was utilized. Furthermore, the conventional dynamic focusing module was modeled as an aberration-free paraxial surface with a variable focal length to represent the ideal Z-axis shift. During the multi-configuration optimization for the 3D scanning volume, the focal length of this paraxial surface was dynamically adjusted within the ranges of (-∞, -1250 mm] and [1450 mm, ∞). This parameter variation precisely corresponds to a physical working distance shift from 196.660 mm to 236.367 mm (centered at 216.342 mm), effectively covering the ± 20 mm Z-axis scanning range. In this model, the dynamic focusing module is idealized as lenses without intrinsic aberrations and is directly integrated with the XY galvanometer scanner and the F-theta telecentric lens. System performance was evaluated via numerical simulations using the Physical Optics Propagation (POP) method with a Gaussian input beam. The simulations focused on the OPD and focusing quality across different Z-axis planes and galvanometer deflection angles. By adjusting the focal length of the dynamic focusing module, a Z-axis dynamic focusing range of ±20 mm was achieved. The normalized pupil coordinates along the X and Y directions are denoted by x_p and y_p, respectively. The simulation results are presented in Fig. 2. Figure 2(a)-(b) illustrate the OPD distributions at different Z-axis positions for the on-axis condition, where (θ_x, θ_y) = (0, 0) and the corresponding scan-field coordinate is (x, y) = (0 mm, 0 mm). As the Z-axis position shifts from Z = 0 mm to Z = ±20 mm, the absolute OPD along the x_p and y_p directions increase monotonically, indicating that larger Z-axis offsets induce greater wavefront deviation, particularly for marginal rays. The OPD profiles along the two pupil directions remain nearly identical at each Z plane, consistent with the system's inherent axial symmetry. Under off-axis field conditions, the OPD becomes markedly asymmetric. For a galvanometer deflection of (θ_x, θ_y) = (0, 7), corresponding to (x, y)= (0 mm, 40 mm), the resulting OPD distributions shown in Fig. 2(c)-(d) reveal that the OPD along the X direction stays below 0.25 waves, whereas the OPD along the Y direction increases significantly with both y_p and Z-axis displacement. A comparable but orthogonally oriented trend is observed for (θ_x, θ_y) = (7, 0), corresponding to (x, y) = (40 mm, 0 mm), as illustrated in Fig. 2(e)-(f): the OPD along Y remains within 0.25 waves, while the OPD along X increases substantially with increasing x_p and Z-axis offset.

Overall, the results in Fig. 2(a)-(f) demonstrate a clear dependence of OPD on both galvanometer deflection and axial focusing position. Within the Z-axis range of |Z| ≤ 10 mm, the OPD remains below 0.5 waves across the full scan field, indicating stable wavefront quality. Beyond this range,

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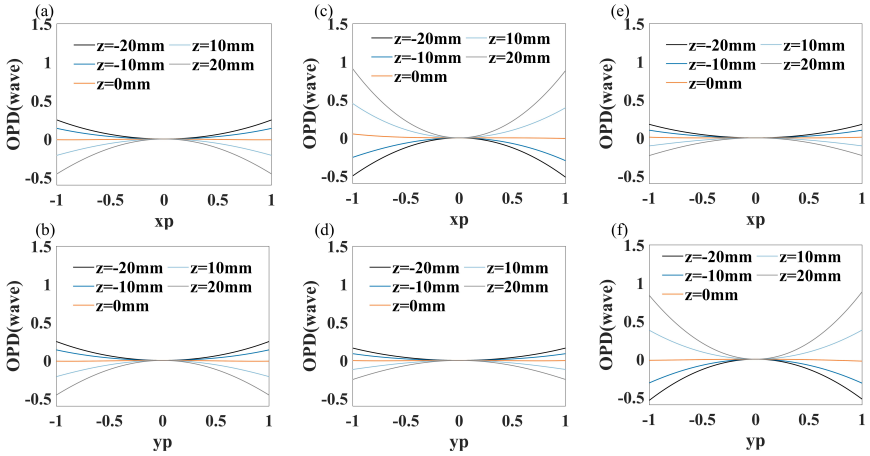

Fig. 2. OPD distribution of the directly integrated transmissive 3D laser scanning system as a function of normalized pupil coordinates x_p, y_p and focal depth Z under different scanning conditions: (a, b) Axial field with mirror deflection angles (θ_x, θ_y) = (0, 0); (c,d) Off-axis field along Y direction with (θ_x, θ_y) =(0, 7); (e, f) Off-axis field along X direction with(θ_x, θ_y) = (7, 0).

the OPD increases significantly, reflecting a degradation in the system's focusing performance and confirming the limitations of directly integrated transmissive configurations.

To quantitatively evaluate the focusing quality and energy concentration of the system under different Z-axis planes and galvanometer deflection conditions, the STRH, focal spot diameter and circularity were employed as evaluation metrics. The simulation results are presented in Fig. 3(a)-(c). At the focal plane Z = 0 mm, the system exhibits near diffraction limited performance within a circular scanning field of 80 mm in diameter: the STRH approaches 1, the focal spot circularity remains close to 1, and the focal spot diameter remains stable at approximately 34 μm. These results demonstrate high focusing quality and strong energy concentration across the scanning field. When the focal plane is offset to Z = ±10 mm, the focusing performance begins to degrade: the STRH decreases to approximately 0.8, the focal-spot circularity drops to 80%, and the spot diameter increases, leading to reduced focal-spot uniformity. Further displacement to Z = ±20 mm results in more pronounced degradation: the STRH falls below 0.8, the circularity decreases to approximately 40%, and the focal-spot diameter reaches a maximum of 80 μm, accompanied by a significant reduction in spot uniformity. Under these conditions, the system is no longer suitable for high-precision laser processing.

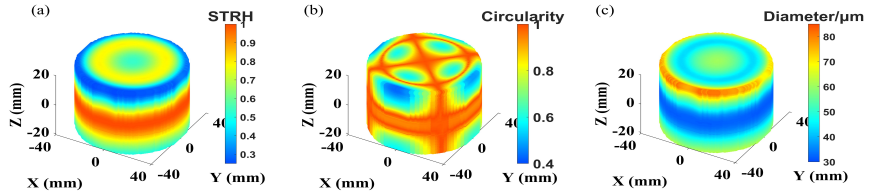

Fig. 3. Focusing performance of the directly integrated transmissive 3D laser scanning system across the 3D scanning range: (a) STRH distribution; (b) Focal spot circularity distribution; (c) Focal spot diameter distribution.

The results in Fig. 2 and Fig. 3 indicate that the effective Z-axis scanning range of a conventional directly integrated transmissive 3D laser scanning system is limited to approximately ±10 mm.

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Beyond this range, system aberrations increase significantly, leading to degradation of the focal spot quality. This limitation arises from the optical design of dynamic focusing module and the F-theta telecentric lens, which are generally designed and optimized independently. During the individual optimization stage, each module minimizes its own OPD under nominal operating conditions. When these modules are integrated into a complete system, optical coupling among the subsystems is neglected, and additional system-level aberrations and OPD are introduced that are not accounted for in the separate designs. During 3D laser scanning, the dynamic focusing module inherently modifies the beam divergence, deviating from the parallel-beam condition required by the F-theta telecentric lens and thereby introducing additional aberrations. Moreover, off-axis beam deflection induced by the XY galvanometer scanner further amplifies the OPD across the scanning field. Under such conditions, the aberrations originating from the dynamic focusing module and the F-theta telecentric lens cannot be effectively compensated. As a result, the focal spot quality progressively degrades along both the axial and lateral directions. This degradation becomes particularly pronounced over large scanning volumes, where key spot metrics—including the STRH, focal spot circularity and diameter—deteriorate significantly. Consequently, the processing uniformity is reduced, the high-quality focusing range is substantially narrowed, and the achievable 3D scanning range of the system is severely limited.

2.3. *Simulation and analysis of the system-level integrated co-optimized transmissive 3D laser scanning system*

To overcome this limitation and further extend the scanning range of the transmissive 3D laser scanning system, a SLICO strategy is proposed. In this strategy, the dynamic focusing module, XY galvanometer scanner, and F-theta telecentric lens are treated as a single, unified optical subsystem. Through multi-parameter coordinated design, system-level aberration balance is achieved, resulting in a fundamental improvement in focal-spot quality during 3D scanning. The systematic workflow of this proposed strategy is detailed in the algorithm flowchart presented in Fig. 4. As illustrated, the co-optimization process is fundamentally conducted in two stages: In the first stage, the dynamic focusing module and the F-theta telecentric lens are independently designed and preliminarily optimized. By adjusting parameters such as lens curvature, element thickness, optical material, and inter-element spacing, each module is optimized to minimize standalone aberrations. In the second stage, a SLICO model encompassing all three components is constructed, and system-level iterative optimization is conducted to achieve coordinated performance enhancement across the complete optical path. During this process, evaluation functions that simultaneously consider multiple Z-axis positions and a range of galvanometer deflection angles are introduced. Key performance metrics, including RMS wavefront error, STRH, and spot circularity, are simultaneously constrained. To achieve a system-level aberration balance, the SLICO strategy transitions to a co-optimization stage driven by a composite merit function (MF) within a multi-configuration environment. The MF is defined as:

$$MF = \sqrt{\frac{\sum w_i (V_i - T_i)^2}{\sum w_i}} \tag{8}$$

where the Strehl ratio (STRH) and spot circularity (evaluated by the ratio of orthogonal Encircled Energy (DENC) using the DIVI operand) are designated as the primary image quality metrics. For both STRH and circularity, target values are set to 1.0 with a weight of 1.0 to ensure near-diffraction-limited focusing and high beam symmetry. Simultaneously, the effective 3D scanning depth is rigorously constrained to a 40 mm range with a weight of 1.0 by calculating the difference (DIFF) in back focal lengths (TTHI) between the marginal configurations. The optimization is executed using the Damped Least Squares (DLS) algorithm for rapid local convergence, followed by Hammer optimization to perform a global search for an optimal aberration balance across

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the entire scanning volume. This SLICO strategy enables global aberration balancing across the entire scanning field and a substantially expanded Z-axis focusing range, ensuring that the system maintains near diffraction limited focusing quality even under large Z-axis excursions.

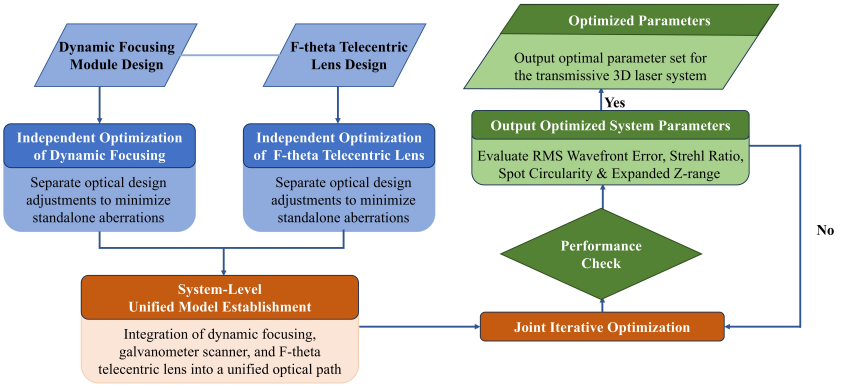

Fig. 4. Algorithm flowchart of the two-stage SLICO strategy.

The SLICO strategy significantly enhances the overall optical performance of the system. As shown in Fig. 5(a)-(f), within a circular XY scanning field of 80 mm in diameter and a Z-axis focusing range of ±20 mm, the optimized system maintains the OPD below 0.25 waves, demonstrating a substantial improvement in wavefront quality. Further analysis of the focal spot metrics, presented in Fig. 6(a)–(c), shows that across this extended 3D scanning range, the STRH remains above 0.96, the focal-spot diameter is stabilized approximately 33.5 μm, and the focal-spot circularity exceeds 98%, closely approaching an ideal circular shape. These results confirm that the focal spot quality is significantly improved compared with the pre-optimized system. Overall, the simulation results verify that the proposed SLICO methodology effectively mitigates aberrations induced by dynamic focusing and XY galvanometer deflection. As a result, the system is capable of maintaining near-diffraction-limited focal performance over a large-area XY scan and extended Z-axis excursions. Compared with the conventional, directly integrated transmissive 3D dynamic focusing system, the optimized design doubles the Z-axis scanning range from ±10 mm to ±20 mm. This improvement demonstrates the effectiveness of the proposed approach in enhancing optical design robustness, improving focusing performance, and expanding the 3D laser processing capability.

To understand the physical mechanisms behind the performance enhancement of the SLICO strategy, a quantitative breakdown of the specific aberrations was performed. Initially, the wavefront errors of the systems were decomposed using standard Zernike polynomials. To provide the most intuitive physical insight into the asymmetric optical paths caused by the inherent physical separation of the X and Y galvanometer mirrors, these low-order Zernike terms were directly mapped to their corresponding Seidel aberrations. These primary aberrations were systematically evaluated at the maximum deflection fields along both orthogonal axes: (θ_x, θ_y) = (0, 7) and (7, 0) under extreme Z-axis shifts (Z = ± 20 mm). The quantitative breakdown is presented in Table 1.

In the conventional integrated system, dynamically adjusting the working distance inherently breaks the aberration balance of the commercial F-theta lens. Off-axis aberrations dominate at the scanning boundaries, particularly astigmatism and field curvature. For example, at (θ_x, θ_y) = (0, 7) with Z = 20 mm, the astigmatism coefficient for the conventional system reaches a substantial value of 0.2309, accompanied by a field curvature of 0.0914. These prominent asymmetrical aberrations inevitably lead to severe focal spot distortion, directly explaining the drastic drop in spot circularity observed in the previous sections.

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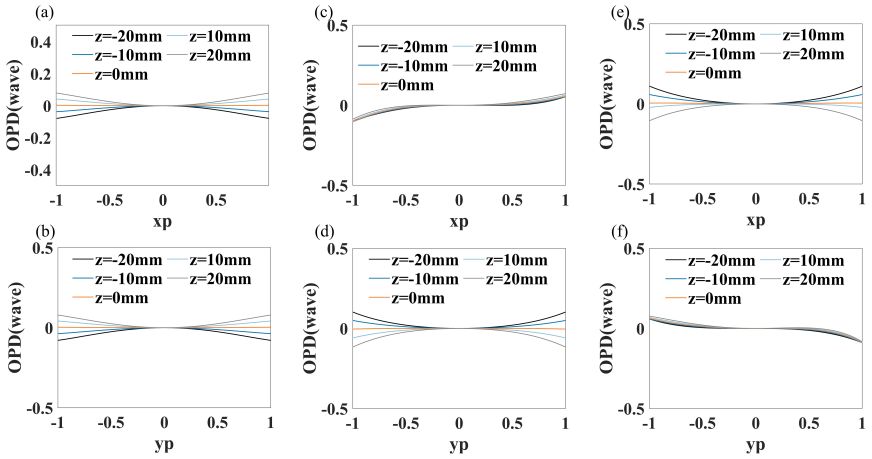

Fig. 5. OPD distribution of the directly integrated transmissive 3D laser scanning system as a function of normalized pupil coordinates x_p, y_p and focal depth Z under different scanning conditions: (a, b) Axial field with mirror deflection angles $(\theta_x, \theta_y) = (0, 0)$; (c,d) Off-axis field along Y direction with $(\theta_x, \theta_y) = (0, 7)$; (e, f) Off-axis field along X direction with$(\theta_x, \theta_y) = (7, 0)$.

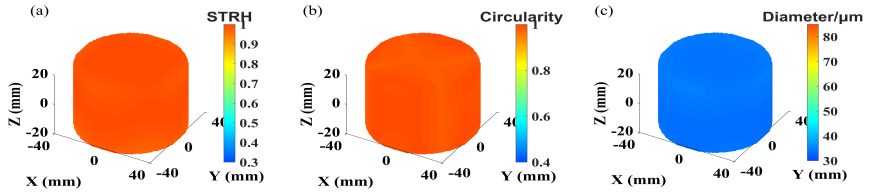

Fig. 6. Focusing performance of the SLICO transmissive 3D laser scanning system across the 3D range: (a) STRH distribution; (b) Focal spot circularity distribution; (c) Focal spot diameter distribution.

Table 1. Quantitative comparison of the primary aberrations (Zernike Standard coefficients, evaluated in waves) between the conventional and optimized SLICO systems. Data are extracted at the maximum orthogonal deflection fields $(\theta_x, \theta_y) = (0, 7)$ and (7, 0) under extreme Z-axis scanning depths (Z = ±20 mm).

	Conventional		SLICO	
Material	z=+20 mm	z = −20 mm	z=+20 mm	z = −20 mm
$(\theta_x, \theta_y) = (0, 7)$				
Field Curvature	0.0914	-0.0506	-0.0169	0.0120
Astigmatism	0.2309	-0.1381	0.0224	-0.0261
Coma	-0.0010	-0.0005	0.0092	0.0087
Spherical Aberration	0.0014	0.0006	-0.0029	0.0020
Distortion	-0.0026	0.0265	0.0265	0.0249
$(\theta_x, \theta_y) = (7, 0)$				
Field Curvature	0.0886	-0.0521	-0.0147	0.0120
Astigmatism	0.2202	0.1463	-0.0222	-0.0261
Coma	0.0036	0.0018	-0.0091	0.0087
Spherical Aberr.	0.0011	0.0005	-0.0040	0.0020
Distortion	0.0105	0.0052	-0.0241	0.0249

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In stark contrast, the proposed SLICO strategy exhibits exceptional capability in aberration balancing. Through global multi-parameter coordination, the SLICO system actively suppresses these critical off-axis aberrations. Under the identical condition (Z = 20 mm, (θ_x, θ_y) = (0, 7)), the astigmatism is dramatically reduced to 0.0224 (an approximate 90.3% reduction), and the field curvature is minimized to -0.0169. Furthermore, other primary aberrations such as spherical aberration and coma are consistently maintained at extremely low levels (near or below 0.03) across all evaluated configurations. This rigorous Zernike breakdown unequivocally proves that the SLICO strategy fundamentally resolves the severe astigmatism induced by large -axis dynamic focusing, thereby ensuring a highly symmetrical, near-diffraction-limited focal spot throughout the expanded 3D scanning volume.

The suppression of these primary off-axis aberrations directly translates to superior contrast transfer capabilities. To comprehensively evaluate this, a Modulation Transfer Function (MTF) analysis was conducted. Figure 7 and Fig. 8 presents the MTF comparisons between the conventional system and the proposed SLICO system at the extreme scanning depths (Z =± 20 mm).

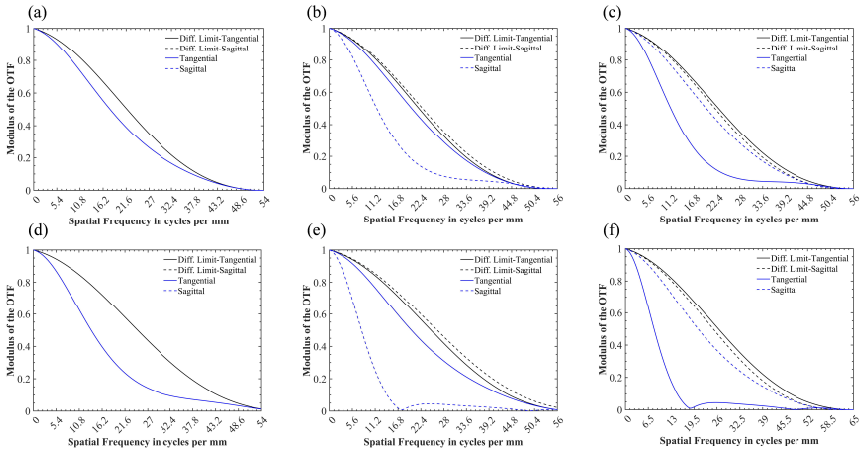

Fig. 7. Modulation Transfer Function (MTF) performance of the conventional reference system evaluated at extreme axial scanning depths. The top row (a–c) shows the MTF curves at a depth of Z = -20 mm, corresponding to field angles of (θx, θy) = (0, 0), (0, 7), and (7, 0), respectively. The bottom row (d–f) presents the corresponding MTF results at a depth of Z = + 20 mm for the same set of field angles. The dashed curves represent the diffraction limit, while the solid lines show the tangential (T) and sagittal (S) MTF response.

As illustrated in the MTF plots, the conventional integrated system suffers from a severe decline in MTF at higher spatial frequencies when deflected to the edge fields. This rapid loss of contrast transfer efficiency aligns with the massive accumulation of primary astigmatism and field curvature observed in Table 1, indicating a significant degradation in focal sharpness at the scanning boundaries. In stark contrast, by globally balancing the off-axis aberrations through the co-optimization of the dynamic focusing module and the F-theta lens, the SLICO system exhibits MTF curves that closely follow the diffraction limit. Notably, this high contrast transfer is maintained uniformly across the center and edge fields, even at the maximum Z =± 20 mm. These comprehensive physical simulations conclusively prove that the SLICO strategy ensures robust, high-resolution focusing quality and stable spatial frequency response throughout the entire extended 3D scanning volume. Based on these theoretical and simulated validations, the subsequent section presents the experimental implementation to verify the actual physical performance of the proposed system.

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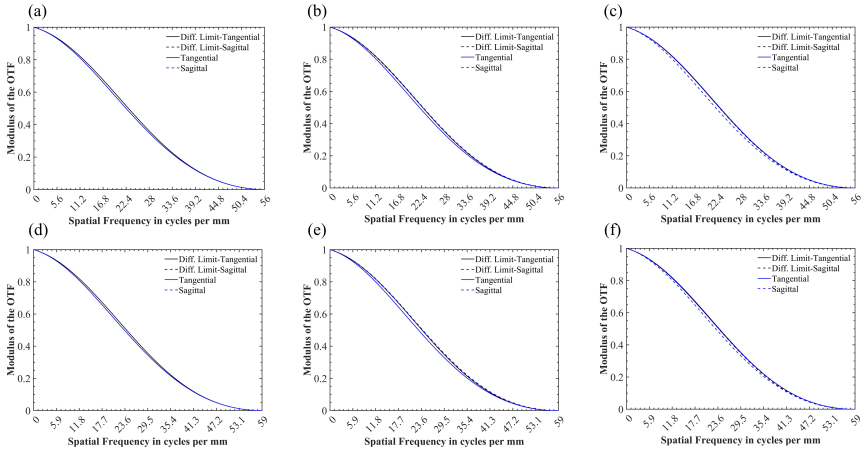

Fig. 8. Modulation Transfer Function (MTF) performance of the SLICO system evaluated at extreme axial scanning depths. The top row (a–c) shows the MTF curves at a depth of Z = -20 mm, corresponding to field angles of (θx, θy) = (0, 0), (0, 7), and (7, 0), respectively. The bottom row (d–f) presents the corresponding MTF results at a depth of Z = + 20 mm for the same set of field angles. The dashed curves represent the diffraction limit, while the solid lines show the tangential (T) and sagittal (S) MTF response.

3. Experimental validation and comparative analysis

Comparative experiments were carried out to verify the effectiveness of the optimized system, and a laser scanning platform incorporating both the conventional directly integrated system and the SLICO transmissive 3D laser scanning system was established, as shown in Fig. 9. A picosecond fiber laser (YDFLP-E2-20-M7-S-R, JPT) with a wavelength of 1064 nm, a beam quality factor of $M^2 = 1.4$ and a maximum average output power of 20 W was employed. To enable a rigorous comparison of scanning dynamics and address the hardware kinematic limits, the specific hardware configurations of both systems are detailed in Table 2. The X- and Y-axis galvanometers, G_1 and G_2, were configured with maximum deflection angles of ±7° for both systems. The focal spot diameter was experimentally determined by analyzing the laser marking results on the sample surface. A Keyence Digital Microscopy System (VHX series) was utilized to capture high-resolution images of the ablation spots and measure their effective diameters. Throughout the measurement trials, the laser power and scanning speed were maintained strictly constant. Consequently, the measured diameter, which is defined by the boundary where the laser intensity exceeds the material's ablation threshold, corresponds closely to the $1/e^2$ beam width. The reported data points represent the arithmetic average of these 3 independent measurements, with the measurement uncertainty expressed as error bars denoting ± 1 standard deviation. Three representative axial planes (Z = 0 mm and Z = ±20 mm) were selected for evaluation. For each plane, four circular scanning paths corresponding to 0.3, 0.5, 0.7, and 1.0 times the full scanning field were defined, with diameters of Φ=24 mm, Φ=40 mm, Φ=56 mm, and Φ=80 mm, respectively. All scans were performed on a conductive-glass substrate ($100 \times 100 \times 1.1$ mm^3). The resulting laser-etched patterns were observed using a laser scanning confocal microscope to quantitatively assess the focal-spot quality, effective scanning range, and scanning stability of both systems.

Figure 10(a-1)-(b-1), (a-2)-(b-2), (e-1)-(f-1), (e-2)-(f-2), (i-1)-(j-1), and (i-2)-(j-2) presented the simulated STRH and focal spot circularities for the conventional directly integrated system and the SLICO transmissive 3D laser scanning system at Z = -20 mm, 0 mm, and 20 mm planes. At the Z = 0 mm plane, both systems achieved STRH exceeding 0.97, with focal spot circularities

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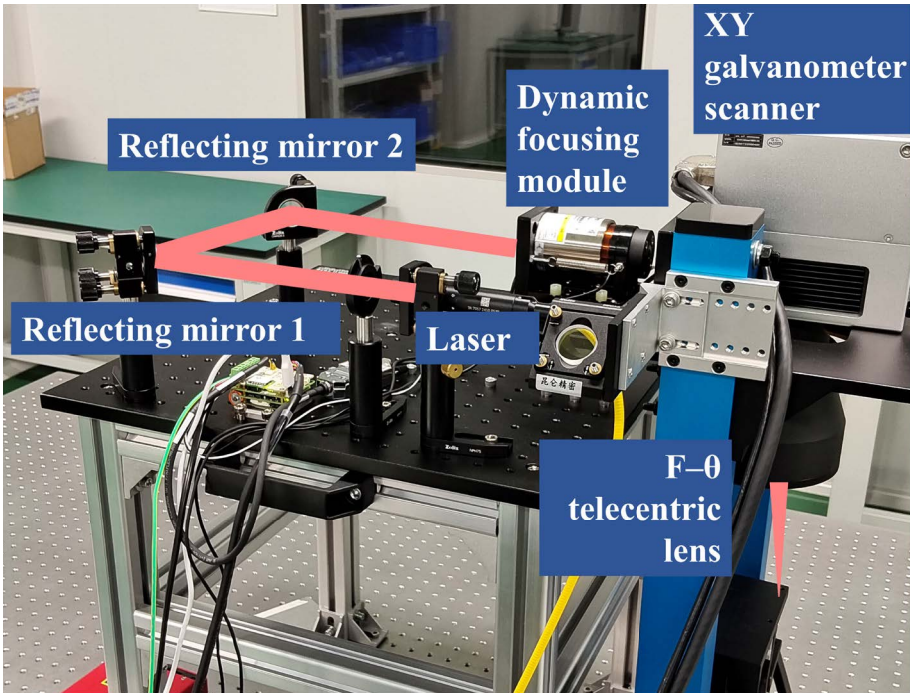

Fig. 9. Experimental setup for the conventional directly integrated and the integrated co-optimized transmissive 3D laser scanning systems.

Table 2. Comparison of detailed hardware configurations and maximum kinematic capabilities between the conventional reference system and the proposed SLCO system

Core Component	Parameter / Function	Traditional Directly Integrated Transmissive System	SLCO Transmissive 3D Laser Scanning System
Dynamic Focusing Module (DFM)	Model / Driver	Scanlab varioSCANde II 20i	Self-developed (Driven by Akribis TGV50 voice coil motor)
	Function	Achieves dynamic laser focusing in the Z-axis	
	Focal Shift Speed	≤ 4.48 m/s	≤ 9 m/s
XY Galvanometer (Scanner)	Model	Scanlab SCANcube IV 14	Optics ExtraScan II 14
	Positioning Speed	14.4 m/s (f = 160 mm)	7 m/s
F-theta Telecentric Lens	Model	LINOS F-Theta-Ronar Lens	Self-developed
	Wavelength	1030 - 1080 nm	1064 nm
	Focal Length (f)	167 mm	167 mm
	Scan Area	85 mm × 85 mm	80 mm × 80 mm

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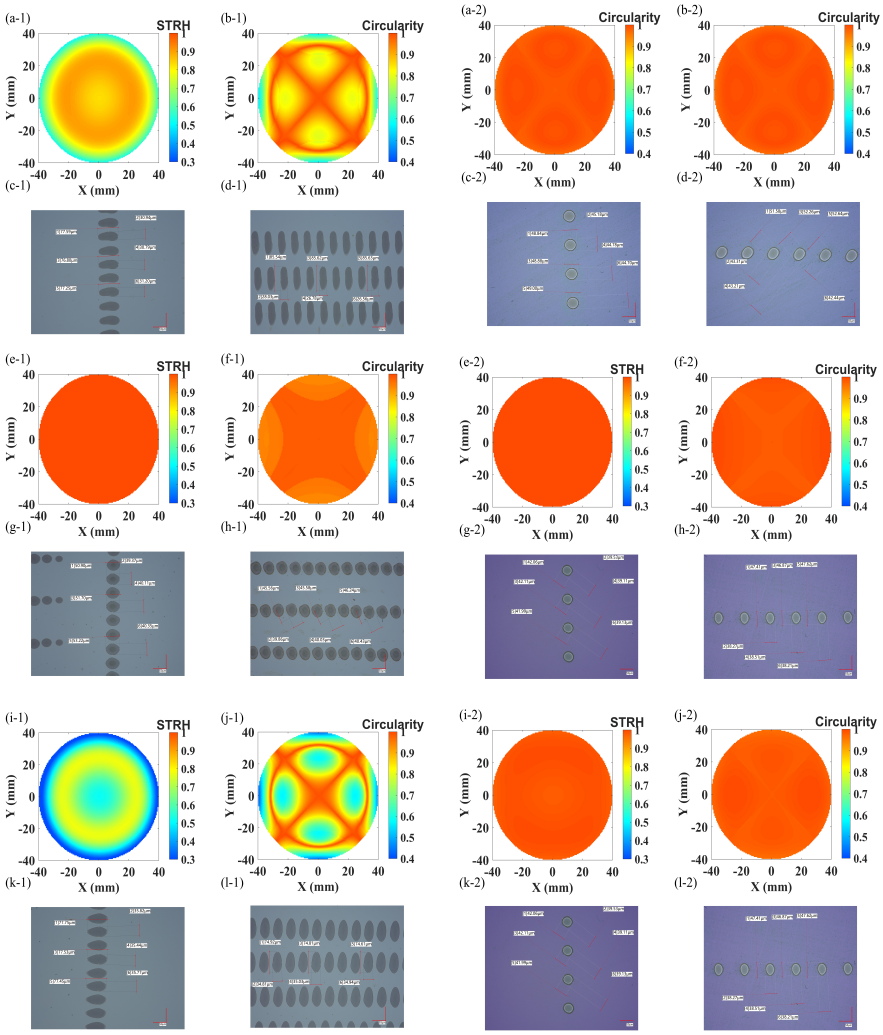

Fig. 10. STRH, focal spot circularity distributions across the full XY plane, and scanning spot images at the outermost field of view for three Z-axis positions. (a-d) Z = -20 mm; (e-h) Z = 0 mm; (i-l) Z = 20 mm. Subscript "-1" corresponds to the traditional directly integrated transmissive 3D laser scanning system, and "-2" corresponds to the SLCO transmissive 3D laser scanning system.

of 97% and 98%, respectively, indicating nearly ideal focusing performance at the central plane. At Z = -20 mm, the conventional system exhibited a STRH of 0.5 and a focal spot circularity of 60%, while at Z = 20 mm, the conventional system further deteriorates, with STRH decreasing to 0.3 and circularity dropping to 50%, particularly at the edges of the scanning field. In contrast, the SLICO system maintained STRH above 0.97 and focal spot circularities exceeding 96.5% across all Z-axis planes, demonstrating robust focusing stability under defocus conditions. The edge-field performance was further evaluated by examining laser-etched spots at the extreme edges of the X and Y scanning fields, as shown in Fig. 10(c-1)-(d-1), (c-2)-(d-2), (g-1)-(h-1), (g-2)-(h-2), (k-1)-(l-1), and (k-2)-(l-2). Quantitative analysis indicates that, for the conventional system, the average edge-spot circularities were 33.3%, 83.4%, and 45.4% at Z = -20 mm, 0 mm,

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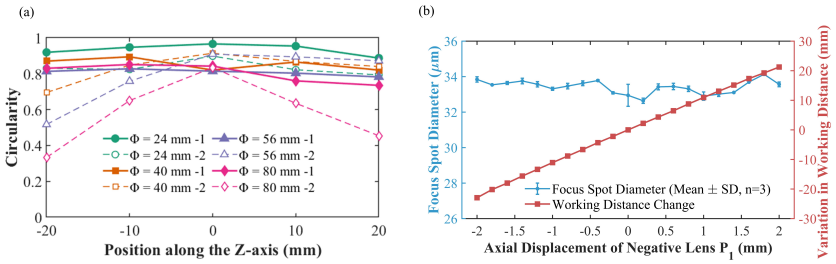

Fig. 11. (a) Experimental results of the focused spot circularity at different Z-axis planes and field positions. The vertical axis (Y-axis) corresponds to field positions of 0.3, 0.5, 0.7, and 1.0 times the full scanning field, representing scanning diameters of Φ = 24 mm, Φ = 40 mm, Φ = 56 mm, and Φ = 80 mm, respectively, used to characterize the focused spot circularity of the system under different field conditions, subscript "-1" corresponds to the traditional directly integrated transmissive 3D laser scanning system, and "-2" corresponds to the SLICO transmissive 3D laser scanning system; (b) Dependence of the focus spot diameter and working distance variation on the axial displacement of the negative lens P1. For the focus spot diameter, each data point represents the arithmetic average of three independent experimental measurements. The error bars indicate ±1 standard deviation (SD).

and 20 mm, respectively. In contrast, the SLICO system achieved 83%, 84.2%, and 73.5% under the same conditions, confirming substantial improvement in edge-field focusing performance and uniformity.

Figure 11(a) further presents the measured circularity of laser-etched spots across multiple scan-field diameters. At Z = 0 mm, both systems exhibited circularities above 80%. For the conventional system, the circularity decreased significantly with increasing Z-axis offset, following simulation predictions. Specifically, at Z=±10 mm, the average circularity of the edge-field laser-etched spots decreased to 63.5% and 65%, respectively, and further declined to 45.4% and 33.3% at Z=±20 mm, effectively limiting the Z-axis scanning range to ±10 mm (Fig. 10(c-1)-(d-1) and Fig. 10(k-1)–(l-1)). In contrast, the SLICO system exhibited only a marginal reduction in circularity with increasing Z-axis offset, maintaining 76% and 85.1% at Z = ±10 mm, and 73.5% and 83% at Z = ±20 mm (Fig. 10(c-2)–(d-2) and Fig. 10(k-2)–(l-2)). Although the conventional system demonstrated marginally higher circularity in 0.3 and 0.5 times of the full scanning field at Z = 0 mm, the co-optimized system consistently maintained circularity above 73.5% across all scanning range, demonstrating a clear performance advantage over the conventional design. The stability of the optimized system was further quantified. Figure 11(b) showed the fluctuation of the focal spot diameter over a Z-axis scanning-induced working distance variation of 44.18 mm. The mean spot diameter was 33.35 μm, with a standard deviation of 0.32 μm, variance of 0.102 μm^2, and a coefficient of variation of 0.96%, demonstrating minimal relative fluctuation. In addition, the measured axial displacement of the negative lens P_1 varied approximately linearly with the system working distance, confirming the accuracy of Eq. (1) and validating the simulation results. Collectively, these results confirm that the SLICO 3D laser scanning system effectively mitigates the degradation of laser-etched spot circularity with increasing Z-axis offset, extending the Z-axis scanning range from ±10 mm to ±20 mm. This demonstrates significant enhancement in focal-spot quality, edge-field uniformity, and overall 3D scanning performance, confirming that the proposed method is for high-precision laser microfabrication with stringent requirements for spot uniformity and machining accuracy.

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4. Conclusion

In conclusion, this study systematically analyzed the 3D scanning mechanism of a transmissive 3D laser scanning system The analysis focused on focal-spot degradation and optical aberrations introduced by the dynamic focusing module, the XY galvanometer scanner, and the F-theta telecentric lens under large-field scanning and Z-axis refocusing conditions. To mitigate the additional system-level aberrations and the limited 3D scanning range caused by large-field angular scanning and dynamic axial focusing, a SLICO strategy was developed, coordinating all optical modules. This method reduces the OPD from 1.0 waves to 0.25 waves, increases the STRH from 0.3 to above 0.96, and stabilizes the focal spot diameter across the 3D scanning range. Experimental results verify that the average circularity of laser-etched spots is ≥73.5%, the relative fluctuation of spot diameter is below 1%, and the effective Z-axis scanning range is doubled from ±10 mm to ±20 mm within a circular XY scanning field of 80 mm in diameter. The results demonstrate that the SLICO strategy effectively suppresses aberrations, maintains high focal spot quality and expands the 3D scanning range. This work not only provides a generalizable methodology for high-precision, large-field 3D laser scanning systems but also offers a practical engineering solution for precision micro- and nanofabrication.

Funding. Shandong Province Technology-Based SME Innovation Enhancement Project (2024TSGC0715); Postgraduate Education Reform Project of Shandong Province (SDYJSJGC2024107); Natural Science Foundation of Shandong Province (Grant No. ZR2021MF122); National Natural Science Foundation of China (Grant No.12204132); Guangdong Provincial Key Research and Development Program (2023B0909010006); Natural Science Foundation of Wuhan (2025040601030123); Wuhan Science and Technology Plan Project.

Disclosures. The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability. Data underlying the results presented in this paper are not publicly available at this time but may be obtained from the authors upon reasonable request.

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